

影像處理 6_視覺神經網路

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Pytorch 訓練網路

```
1  # Train Network
2  model.train()
3  for epoch in range(1):
4      for batch_idx, (data, targets) in enumerate(tqdm(train_loader)):
5          # Get data to cuda if possible
6          data = data.to("cuda")
7          targets = targets.to("cuda")
8
9          # forward
10         scores = model(data)
11
12         loss = criterion(scores, targets)
13
14         # backward
15         optimizer.zero_grad()
16         loss.backward()
17
18         # gradient descent or adam step
19         optimizer.step()
```

作用?

Backpropagation

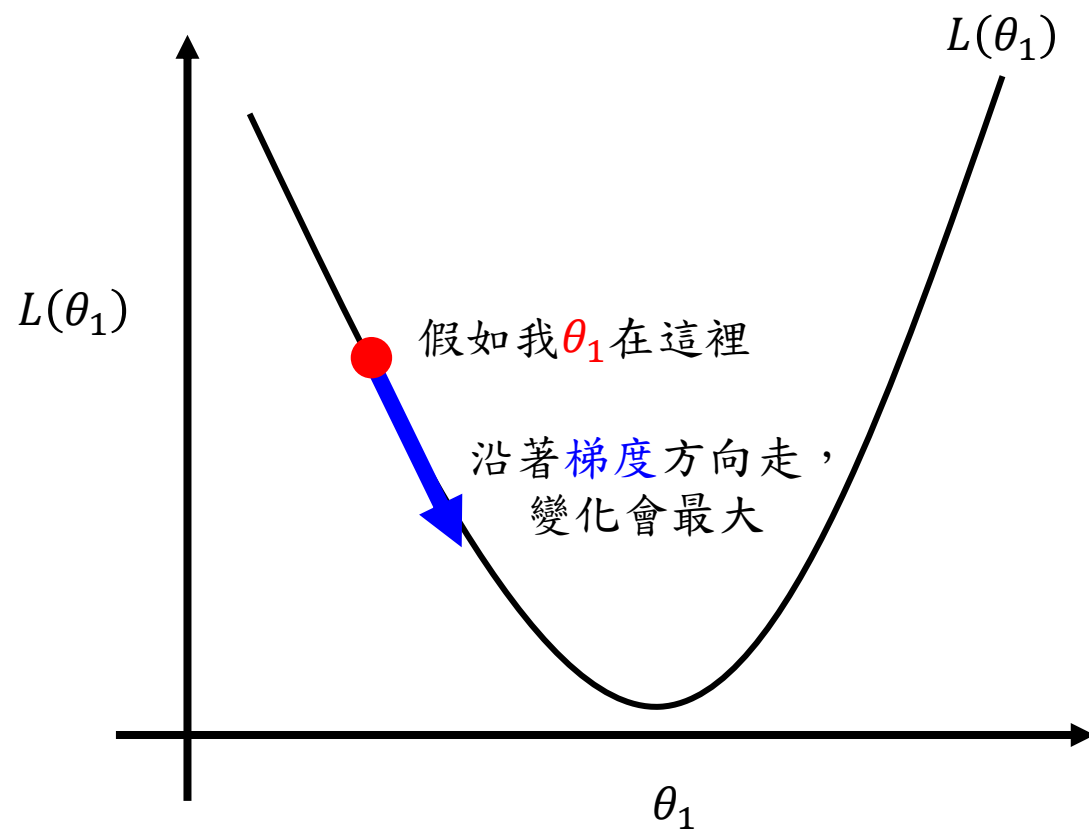
倒傳遞

什麼是Back propagation

- 更新權重的技巧
- 使用Chain Rule(鏈鎖律)達成
- 1970年Seppo Linnainmaa 發表反向傳播
- 1974 年Paul J. Werbos 將反向傳播用在神經網路訓練
- 1985 年Geoffrey Hinton 推廣並證明倒傳遞可以學習特徵



Gradient descent 複習



權重更新方法: $\theta'_1 = \theta_1 - \eta \nabla L(\theta_1)$

新權重=舊權重-學習速率*梯度方向

當中 $\nabla L(\theta_1)$ 又可以寫成 $\frac{\partial L(\theta_1)}{\partial \theta_1}$

亦即 對函數中的**權重**取微分即可得到**梯度**

θ'_1 : 新的參數

θ_1 : 舊參數

η : 學習速率 learning rate

∇ : 梯度

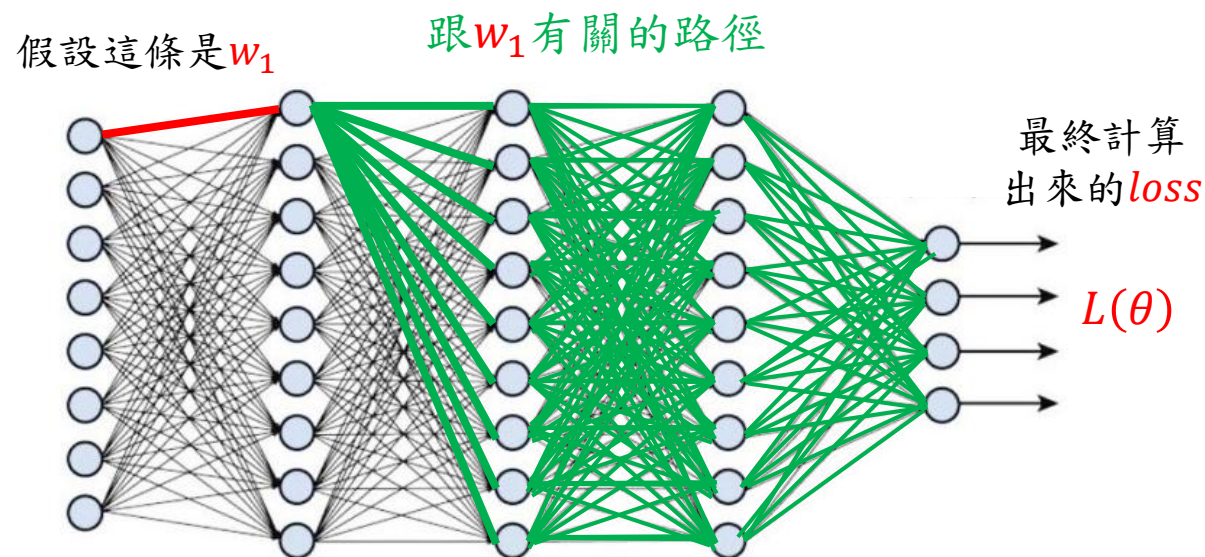
$L()$: 損失函數 loss function

參數很多的時

假如: $\theta = \{w_1, w_2, w_3, w_4, \dots\}$

$$\nabla L(\theta) = \begin{bmatrix} \partial L(\theta) / \partial w_1 \\ \partial L(\theta) / \partial w_2 \\ \partial L(\theta) / \partial w_3 \\ \partial L(\theta) / \partial w_4 \\ \vdots \\ \vdots \end{bmatrix}$$

對每個權重都進行微分



可以看到如果要計算第一項 $\frac{\partial L(\theta)}{\partial w_1}$ 就會很複雜了

因為 w_1 並不是直接影響 $L(\theta)$ ，而是間接影響

Chain Rule複習

Case 1:

$$y = g(x) \quad z = h(y)$$

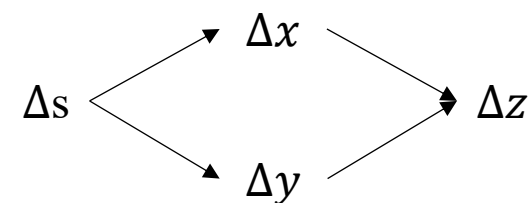
$$\Delta x \rightarrow \Delta y \rightarrow \Delta z$$

改變 x 就會改變 y 就會改變 z

所以這個case的 $\frac{dz}{dx} = \frac{dz}{dy} \frac{dy}{dx}$

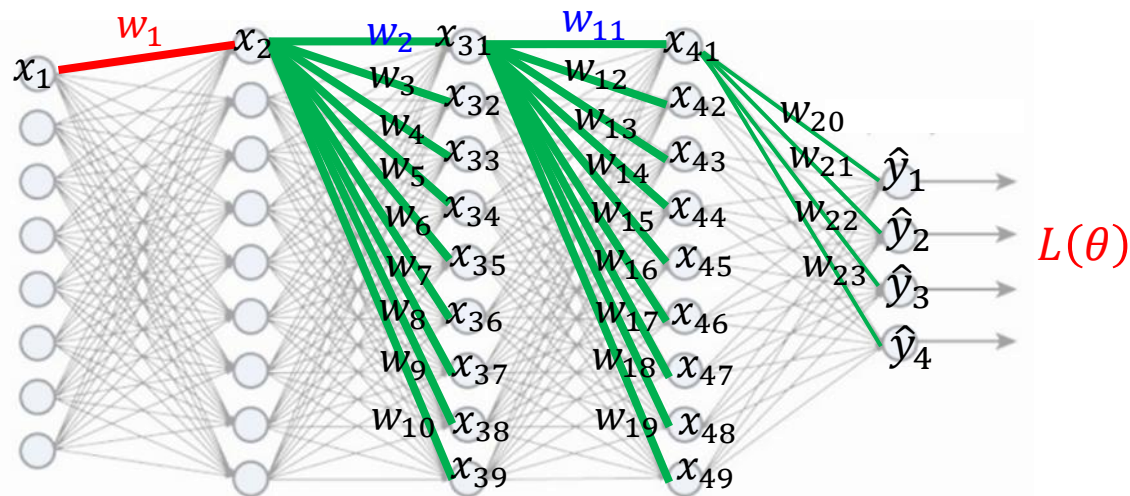
Case 2:

$$x = g(s) \quad y = h(s) \quad z = k(x, y)$$



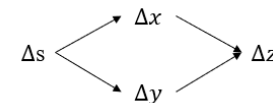
改變 s ，會同時改變 x 跟 y ，就會改變 z

所以這個case的 $\frac{dz}{ds} = \frac{dz}{dx} \frac{dx}{ds} + \frac{dz}{dy} \frac{dy}{ds}$



Case 2:

$$x = g(s) \quad y = h(s) \quad z = k(x, y)$$



改變 s ，會同時改變 x 跟 y ，就會改變 z

$$\text{所以這個case的 } \frac{dz}{ds} = \frac{dz}{dx} \frac{dx}{ds} + \frac{dz}{dy} \frac{dy}{ds}$$

今天要算 $\frac{\partial L(\theta)}{\partial w_1} = \frac{\partial L(\theta)}{\partial x_2} \frac{\partial x_2}{\partial w_1}$

“假設都沒有 activation func 跟 bias”

$$\frac{\partial L(\theta)}{\partial x_2} = \frac{\partial L(\theta)}{\partial x_{31}} \frac{\partial x_{31}}{\partial x_2} + \frac{\partial L(\theta)}{\partial x_{32}} \frac{\partial x_{32}}{\partial x_2} + \dots + \frac{\partial L(\theta)}{\partial x_{39}} \frac{\partial x_{39}}{\partial x_2}$$

$$\frac{\partial L(\theta)}{\partial x_{31}} = \frac{\partial L(\theta)}{\partial x_{41}} \frac{\partial x_{41}}{\partial x_{31}} + \frac{\partial L(\theta)}{\partial x_{42}} \frac{\partial x_{42}}{\partial x_{31}} + \dots + \frac{\partial L(\theta)}{\partial x_{49}} \frac{\partial x_{49}}{\partial x_{31}}$$

$$\frac{\partial L(\theta)}{\partial x_{41}} = \frac{\partial L(\theta)}{\partial y_1} \frac{\partial y_1}{\partial x_{41}} + \frac{\partial L(\theta)}{\partial y_2} \frac{\partial y_2}{\partial x_{41}} + \frac{\partial L(\theta)}{\partial y_3} \frac{\partial y_3}{\partial x_{41}} + \frac{\partial L(\theta)}{\partial y_4} \frac{\partial y_4}{\partial x_{41}}$$

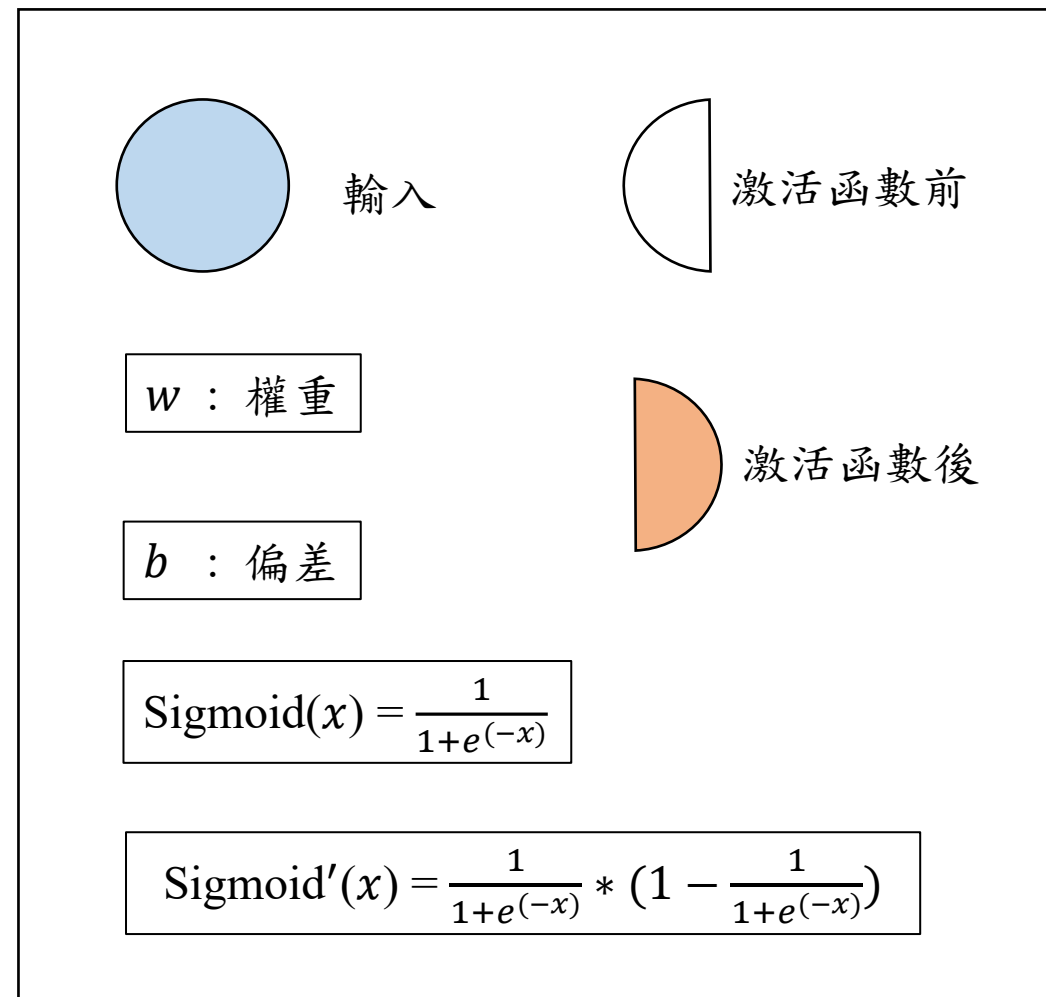
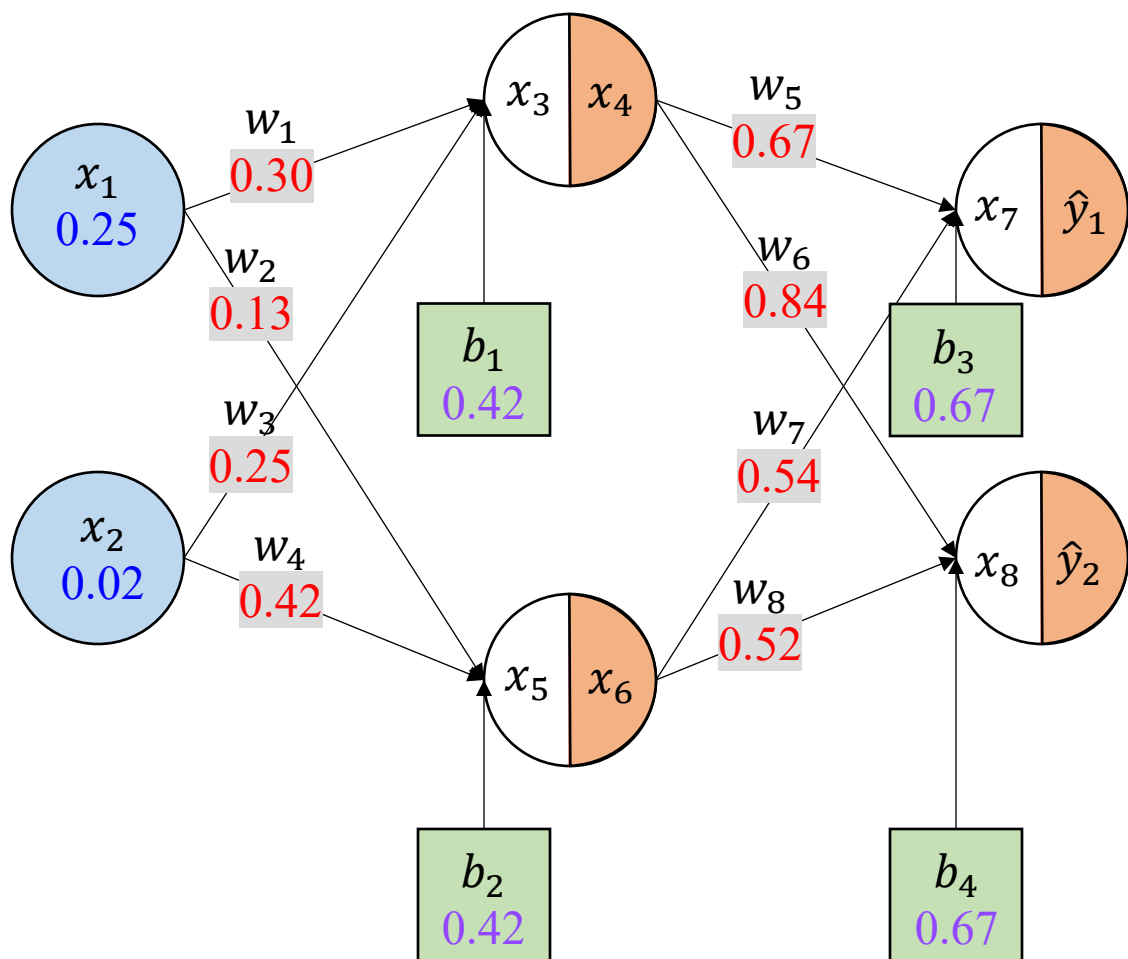
拆到最後 $\frac{\partial L(\theta)}{\partial y_1} = \frac{\partial [\frac{1}{2}(y_1 - \hat{y}_1)^2]}{\partial (y_1 - \hat{y}_1)} \frac{\partial (y_1 - \hat{y}_1)}{\partial \hat{y}_1} = (y_1 - \hat{y}_1) * (-1) = (\hat{y}_1 - y_1)$

$$L(\theta) = \frac{1}{2}(y_1 - \hat{y}_1)^2 + \frac{1}{2}(y_2 - \hat{y}_2)^2 + \frac{1}{2}(y_3 - \hat{y}_3)^2 + \frac{1}{2}(y_4 - \hat{y}_4)^2$$

$$\hat{y}_1 = x_4 w_{20} + \dots + x_n w_n$$

結論:逆著算比較沒那麼多未知數

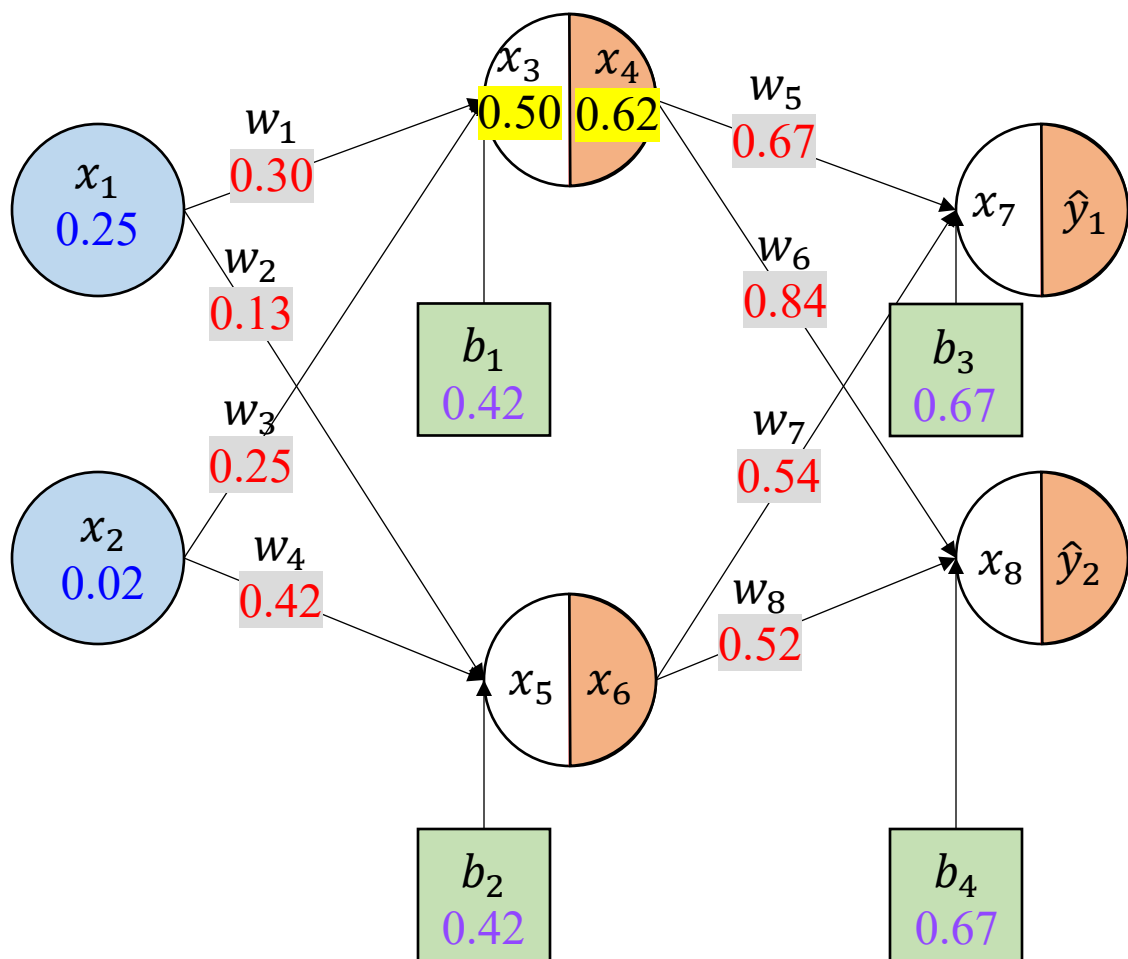
範例



Forward pass

計算每顆神經元的值

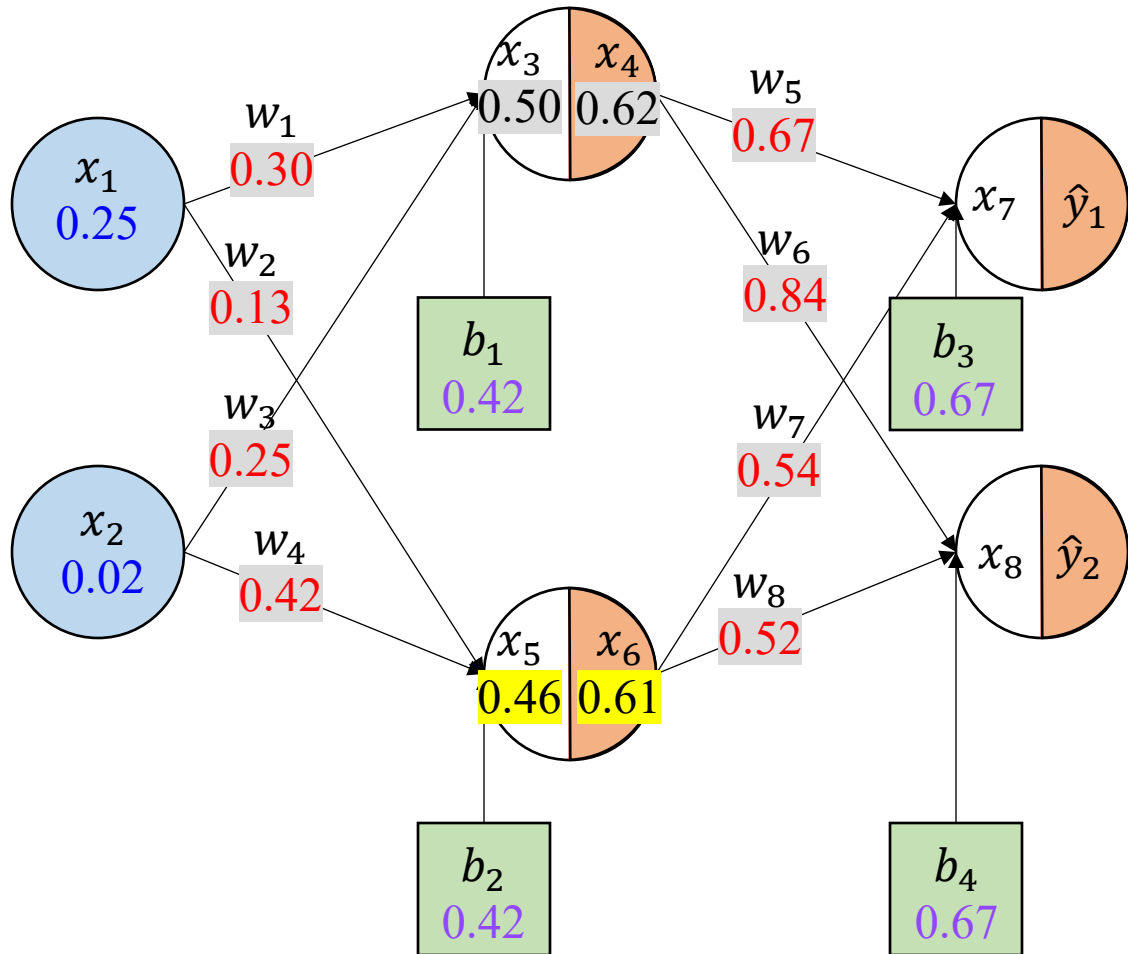
範例(Forward)



$$\begin{aligned}x_3 &= x_1 w_1 + x_2 w_3 + b_1 \\&= 0.25 * 0.30 + 0.02 * 0.25 + 0.42 \\&= 0.5\end{aligned}$$

$$\begin{aligned}x_4 &= \text{Sigmoid}(x_3) \\&= \frac{1}{1+e^{(-x_3)}} \\&= \frac{1}{1+e^{(-0.5)}} \\&= 0.62\end{aligned}$$

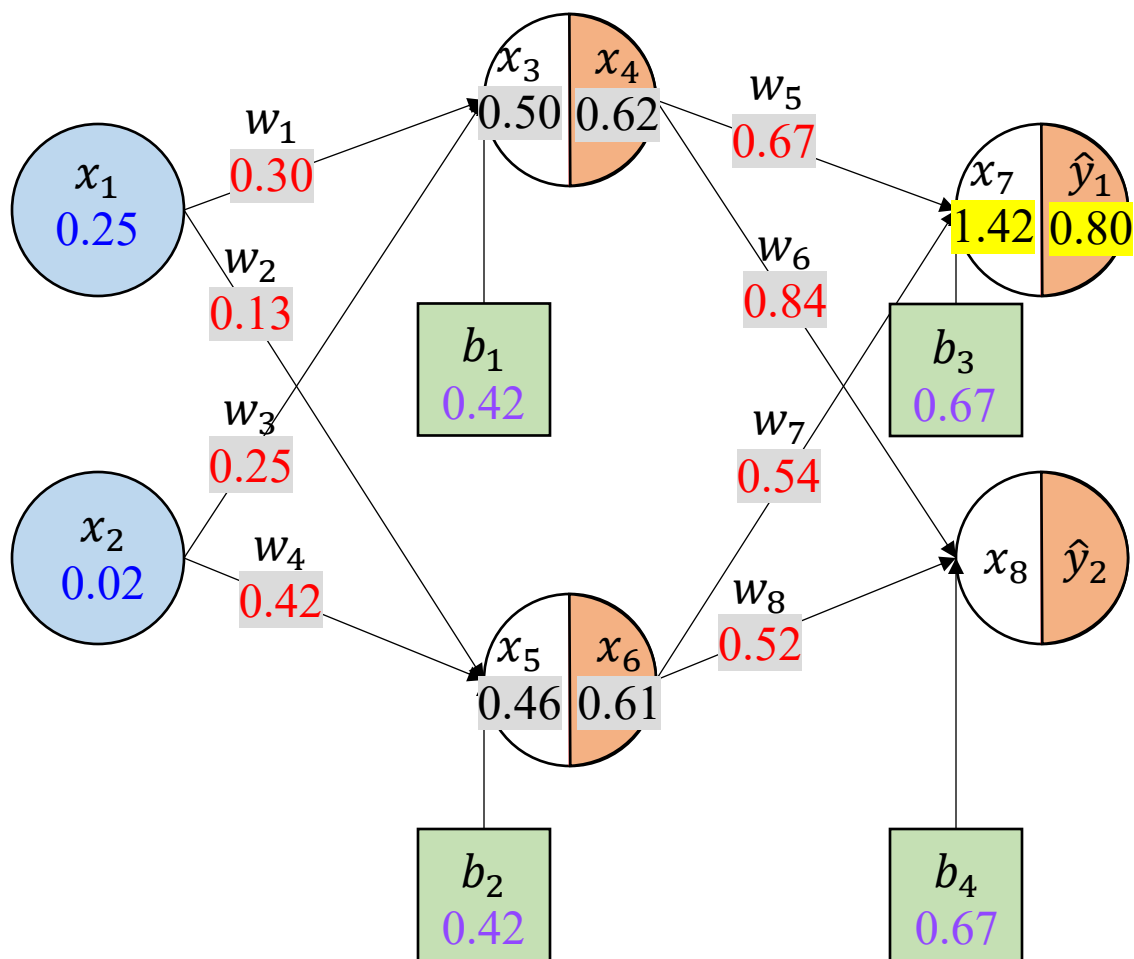
範例(Forward)



$$\begin{aligned}x_5 &= x_1 w_2 + x_2 w_4 + b_2 \\&= 0.25 * 0.13 + 0.02 * 0.42 + 0.42 \\&= 0.46\end{aligned}$$

$$\begin{aligned}x_6 &= \text{Sigmoid}(x_5) \\&= \frac{1}{1 + e^{(-0.46)}} \\&= 0.61\end{aligned}$$

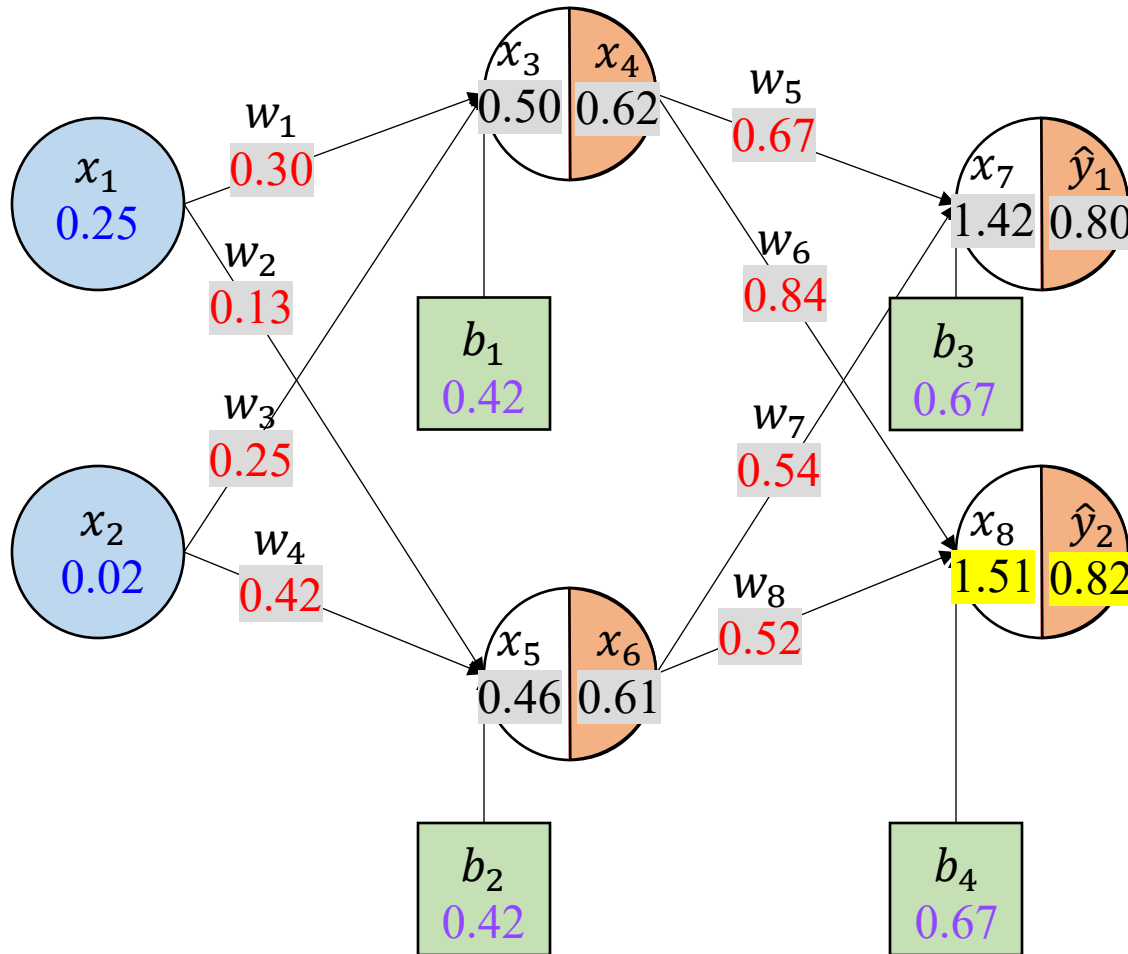
範例(Forward)



$$\begin{aligned}
 x_7 &= x_4 w_5 + x_6 w_7 + b_3 \\
 &= 0.62 * 0.67 + 0.61 * 0.54 + 0.67 \\
 &= 1.42
 \end{aligned}$$

$$\begin{aligned}
 \hat{y}_1 &= \frac{1}{1 + e^{-x_7}} \\
 &= \frac{1}{1 + e^{-1.42}} \\
 &= 0.80
 \end{aligned}$$

範例(Forward)

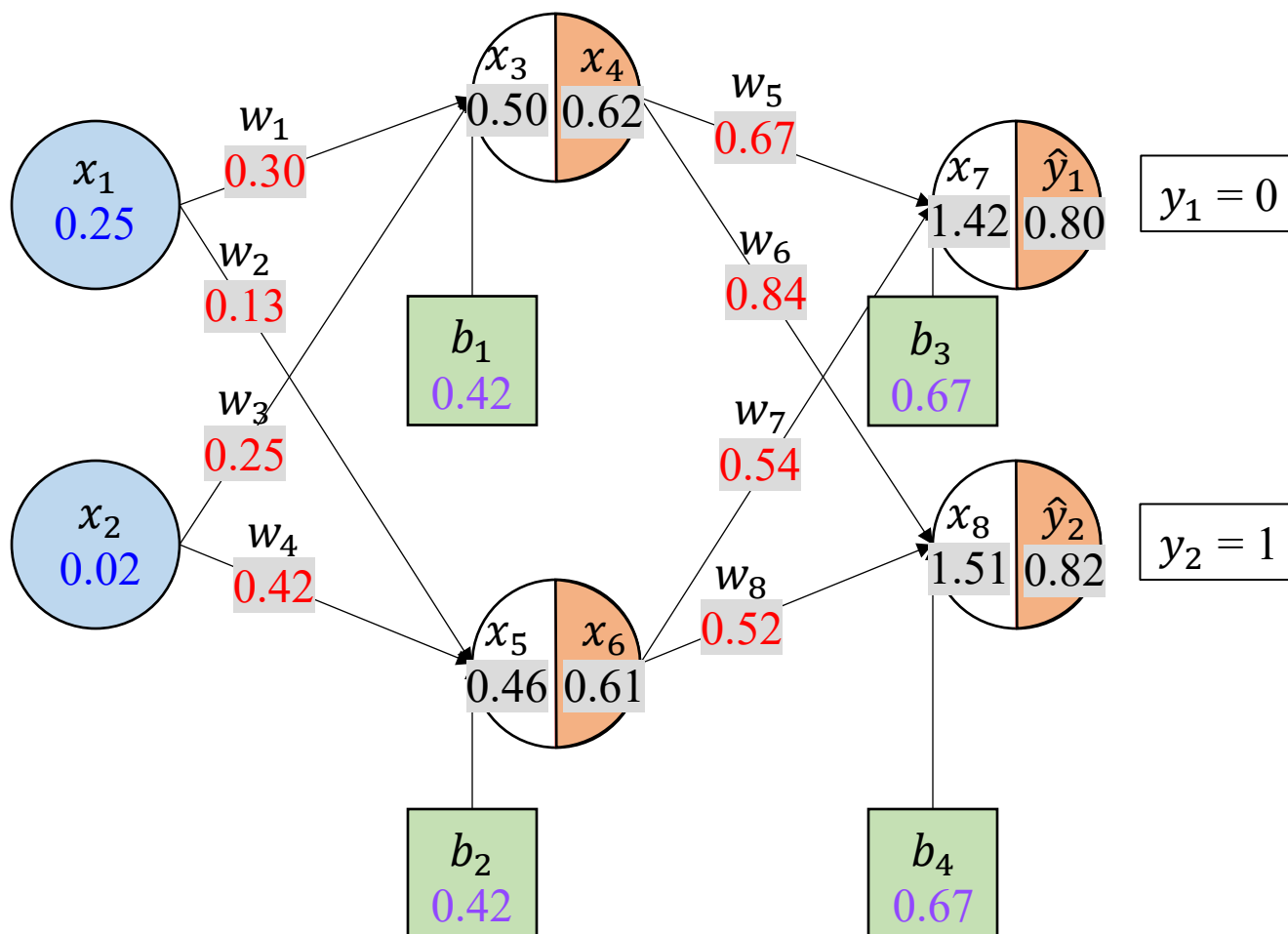


$$\begin{aligned}
 x_8 &= x_4 w_6 + x_6 w_8 + b_4 \\
 &= 0.62 * 0.84 + 0.61 * 0.52 + 0.67 \\
 &= 1.51
 \end{aligned}$$

$$\begin{aligned}
 \hat{y}_2 &= \frac{1}{1 + e^{(-x_8)}} \\
 &= \frac{1}{1 + e^{(-1.51)}} \\
 &= 0.82
 \end{aligned}$$

計算損失 L_{total}

範例(計算 L_{total})



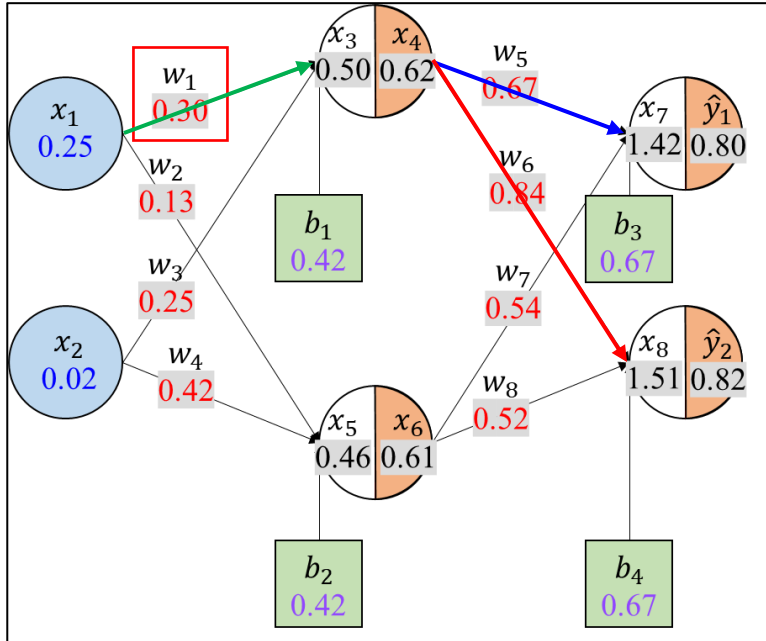
$$L_{y_1} = \frac{1}{2} (y_1 - \hat{y}_1)^2 = \frac{1}{2} (0 - 0.80)^2 = 0.32$$

$$L_{y_2} = \frac{1}{2} (y_2 - \hat{y}_2)^2 = \frac{1}{2} (1 - 0.82)^2 = 0.0162$$

$$L_{\text{total}} = L_{y_1} + L_{y_2} = 0.3362$$

$$L_{\text{total}} = \frac{1}{2} (y_1 - \hat{y}_1)^2 + \frac{1}{2} (y_2 - \hat{y}_2)^2$$

1. Loss 對輸出的偏微分 $\frac{\partial L_{total}}{\partial w_1}$



w_1 只影響隱藏層的 $x_3 \rightarrow x_4$ ，再往右影響兩個輸出 $\hat{y}_1, \hat{y}_2$ 。
所以：

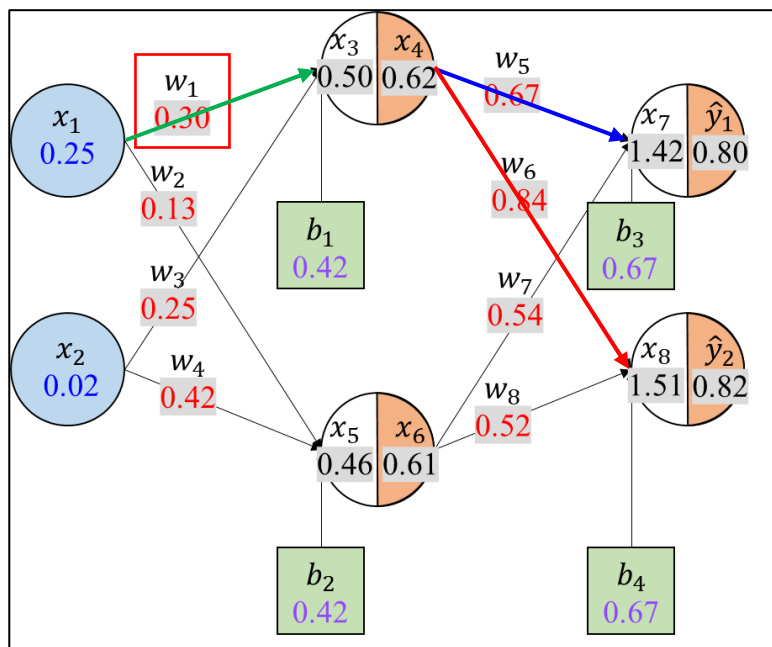
$$\frac{\partial L_{total}}{\partial w_1} = \left(\frac{\partial L_{total}}{\partial \hat{y}_1} \frac{\partial \hat{y}_1}{\partial x_7} \frac{\partial x_7}{\partial x_4} + \frac{\partial L_{total}}{\partial \hat{y}_2} \frac{\partial \hat{y}_2}{\partial x_8} \frac{\partial x_8}{\partial x_4} \right) \frac{\partial x_4}{\partial x_3} \frac{\partial x_3}{\partial w_1}$$

$$L_{total} = \frac{1}{2} (y_1 - \hat{y}_1)^2 + \frac{1}{2} (y_2 - \hat{y}_2)^2$$

$$\frac{\partial L_{total}}{\partial \hat{y}_1} = (\hat{y}_1 - y_1)$$

$$\frac{\partial L_{total}}{\partial \hat{y}_2} = (\hat{y}_2 - y_2)$$

2.輸出層的激活函數微分



w_1 只影響隱藏層的 $x_3 \rightarrow x_4$ ，再往右影響兩個輸出 $\hat{y}_1, \hat{y}_2$ 。
所以：

$$\frac{\partial L_{total}}{\partial w_1} = \left(\frac{\partial L_{total}}{\partial \hat{y}_1} \frac{\partial \hat{y}_1}{\partial x_7} \frac{\partial x_7}{\partial x_4} + \frac{\partial L_{total}}{\partial \hat{y}_2} \frac{\partial \hat{y}_2}{\partial x_8} \frac{\partial x_8}{\partial x_4} \right) \frac{\partial x_4}{\partial x_3} \frac{\partial x_3}{\partial w_1}$$

輸出層的激活函數微分 (Sigmoid, $f(x)$)

因為：

$$f(x_7) = \hat{y}_1, \quad f(x_8) = \hat{y}_2$$

sigmoid 導數：

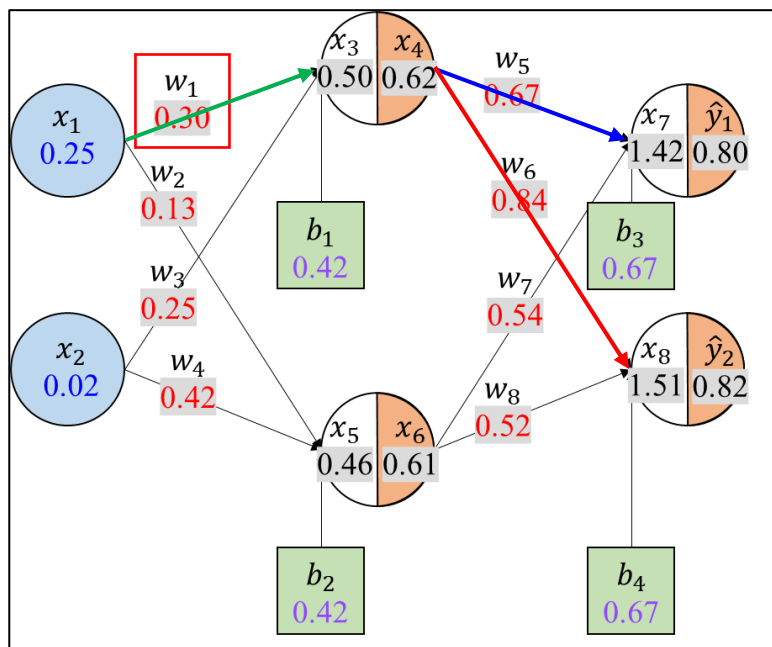
$$f'(x) = f(x) (1 - f(x))$$

所以：

$$\frac{\partial \hat{y}_1}{\partial x_7} = \hat{y}_1 (1 - \hat{y}_1)$$

$$\frac{\partial \hat{y}_2}{\partial x_8} = \hat{y}_2 (1 - \hat{y}_2)$$

3. 線性組合對 x_4 的導數



w_1 只影響隱藏層的 $x_3 \rightarrow x_4$ ，再往右影響兩個輸出 $\hat{y}_1, \hat{y}_2$ 。
所以：

$$\frac{\partial L_{total}}{\partial w_1} = \left(\frac{\partial L_{total}}{\partial \hat{y}_1} \frac{\partial \hat{y}_1}{\partial x_7} \frac{\partial x_7}{\partial x_4} + \frac{\partial L_{total}}{\partial \hat{y}_2} \frac{\partial \hat{y}_2}{\partial x_8} \frac{\partial x_8}{\partial x_4} \right) \frac{\partial x_4}{\partial x_3} \frac{\partial x_3}{\partial w_1}$$

線性組合對 x_4 的導數

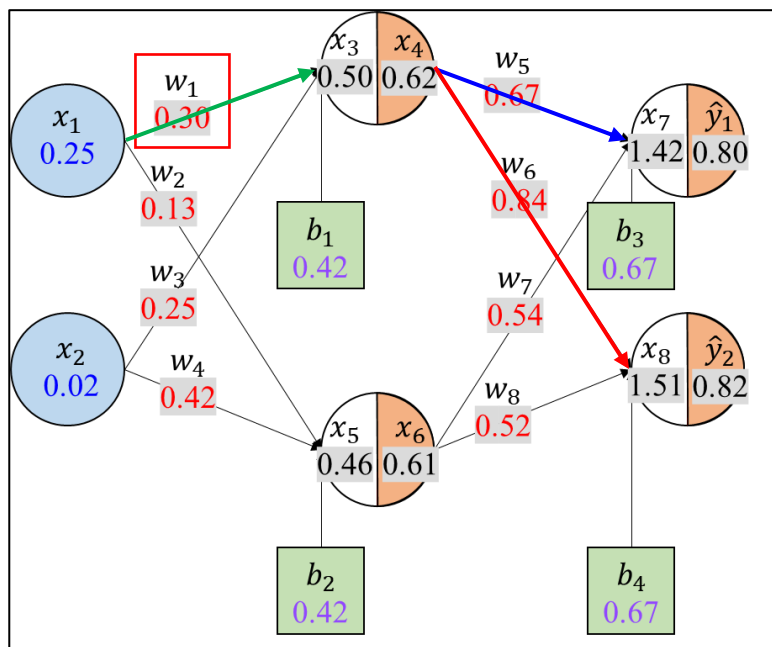
$$x_7 = w_5 x_4 + w_7 x_6 + b_3 \Rightarrow \frac{\partial x_7}{\partial x_4} = w_5$$

$$x_8 = w_6 x_4 + w_8 x_6 + b_4 \Rightarrow \frac{\partial x_8}{\partial x_4} = w_6$$

這三個合併：

$$\frac{\partial L_{total}}{\partial x_4} = (\hat{y}_1 - y_1) \hat{y}_1 (1 - \hat{y}_1) w_5 + (\hat{y}_2 - y_2) \hat{y}_2 (1 - \hat{y}_2) w_6$$

4. 隱藏層的激活函數對 x_3 的導數



w_1 只影響隱藏層的 $x_3 \rightarrow x_4$ ，再往右影響兩個輸出 $\hat{y}_1, \hat{y}_2$ 。
所以：

$$\frac{\partial L_{total}}{\partial w_1} = \left(\frac{\partial L_{total}}{\partial \hat{y}_1} \frac{\partial \hat{y}_1}{\partial x_7} \frac{\partial x_7}{\partial x_4} + \frac{\partial L_{total}}{\partial \hat{y}_2} \frac{\partial \hat{y}_2}{\partial x_8} \frac{\partial x_8}{\partial x_4} \right) \frac{\partial x_4}{\partial x_3} \frac{\partial x_3}{\partial w_1}$$

輸出層的激活函數微分 (Sigmoid, $f(x)$)

因為：

$$f(x_3) = x_4$$

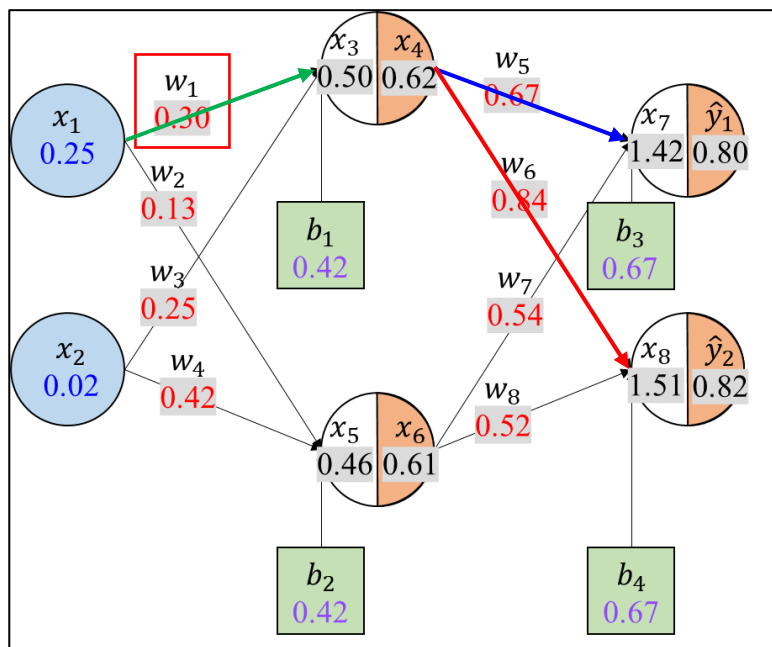
sigmoid 導數：

$$f'(x) = f(x) (1 - f(x))$$

所以：

$$\frac{\partial x_4}{\partial x_3} = f'(x_3) = x_4(1 - x_4)$$

5. 線性組合對 w_1 的導數



w_1 只影響隱藏層的 $x_3 \rightarrow x_4$ ，再往右影響兩個輸出 $\hat{y}_1, \hat{y}_2$ 。
所以：

$$\frac{\partial L_{total}}{\partial w_1} = \left(\frac{\partial L_{total}}{\partial \hat{y}_1} \frac{\partial \hat{y}_1}{\partial x_7} \frac{\partial x_7}{\partial x_4} + \frac{\partial L_{total}}{\partial \hat{y}_2} \frac{\partial \hat{y}_2}{\partial x_8} \frac{\partial x_8}{\partial x_4} \right) \frac{\partial x_4}{\partial x_3} \frac{\partial x_3}{\partial w_1}$$

回到隱藏層的線性組合：

$$x_3 = w_1 x_1 + w_3 x_2 + b_1$$

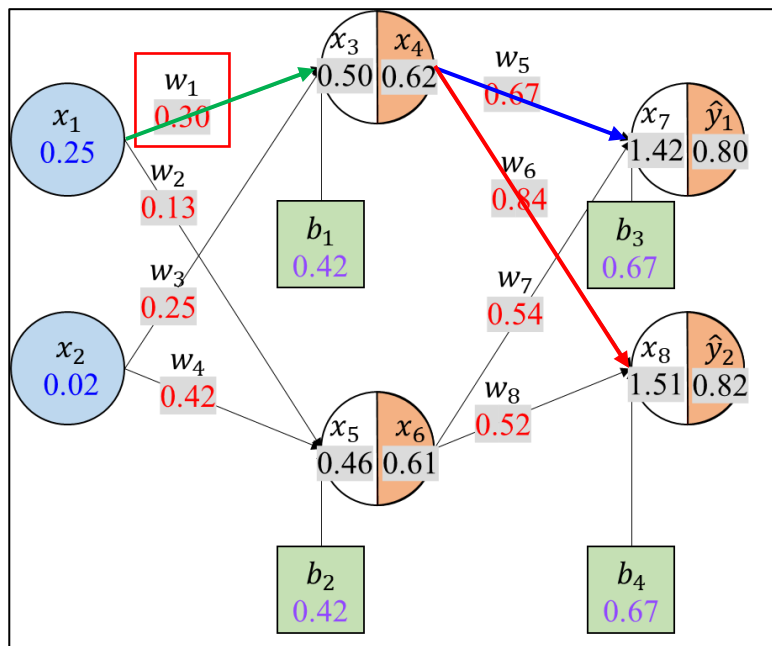
只有第一項跟 w_1 有關：

$$\frac{\partial (w_1 x_1)}{\partial w_1} = x_1$$

所以：

$$\boxed{\frac{\partial x_3}{\partial w_1} = x_1}$$

1~5大合併



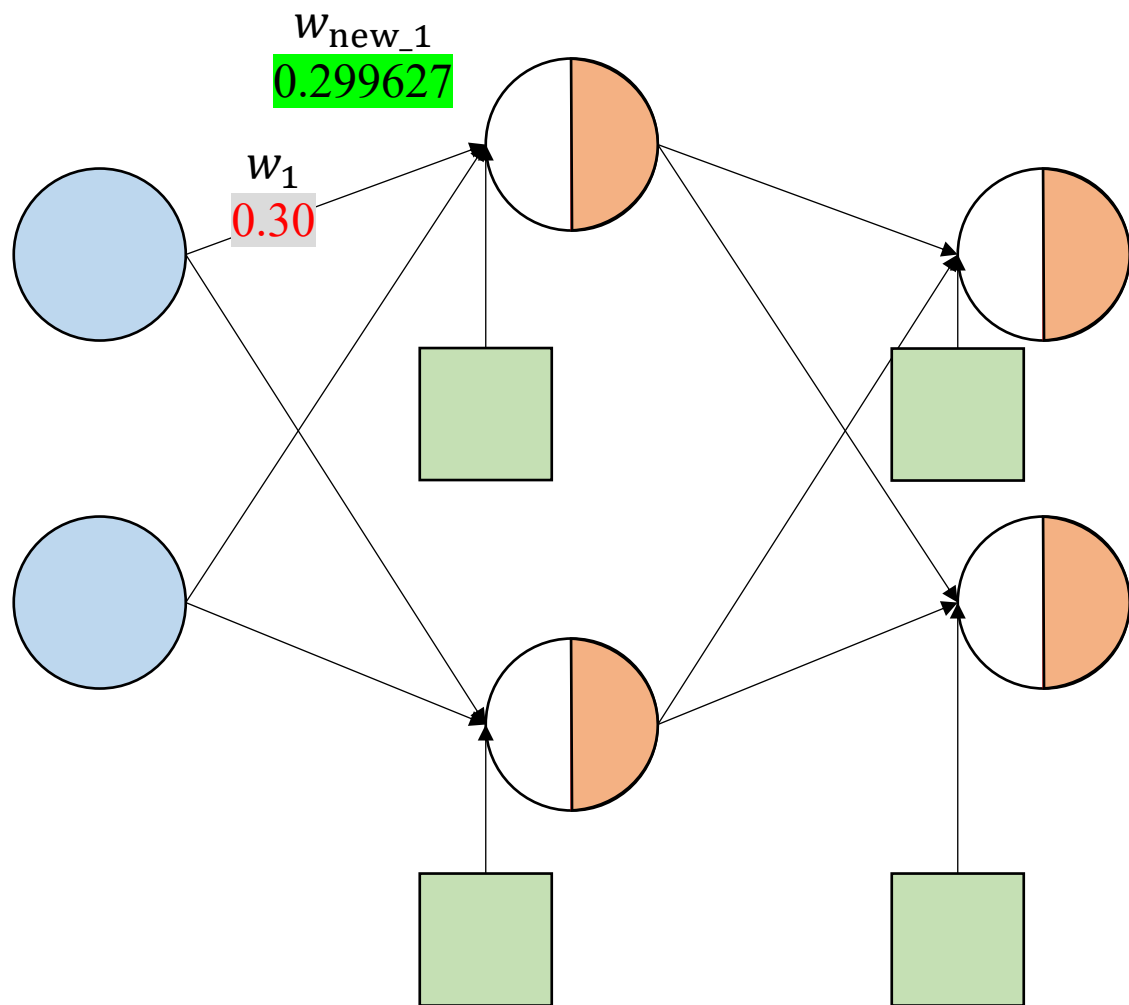
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假設 $y_1 = 0, y_2 = 1$

$$\begin{aligned} \frac{\partial L_{total}}{\partial w_1} &= [(\hat{y}_1 - y_1)\hat{y}_1(1 - \hat{y}_1) w_5 + (\hat{y}_2 - y_2)\hat{y}_2(1 - \hat{y}_2) w_6] x_4(1 - x_4) x_1 \\ &= 0.00373 \end{aligned}$$

更新權重 new w_1



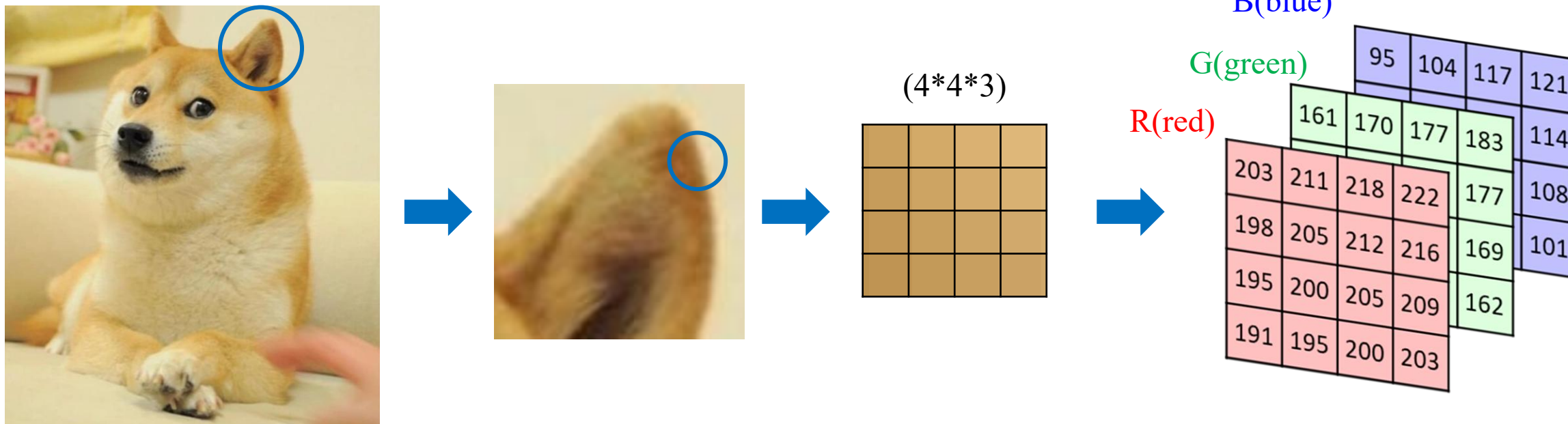
權重更新方式(優化器:SGD)
 $\theta'_1 = \theta_1 - \eta \nabla L(\theta_1)$ 設 $\eta=0.1$

$$\begin{aligned} w_{\text{new}_1} &= w_1 - \eta * \frac{\partial L_{\text{total}}}{\partial w_1} \\ &= 0.30 - 0.1 * (0.00373) = 0.299627 \end{aligned}$$

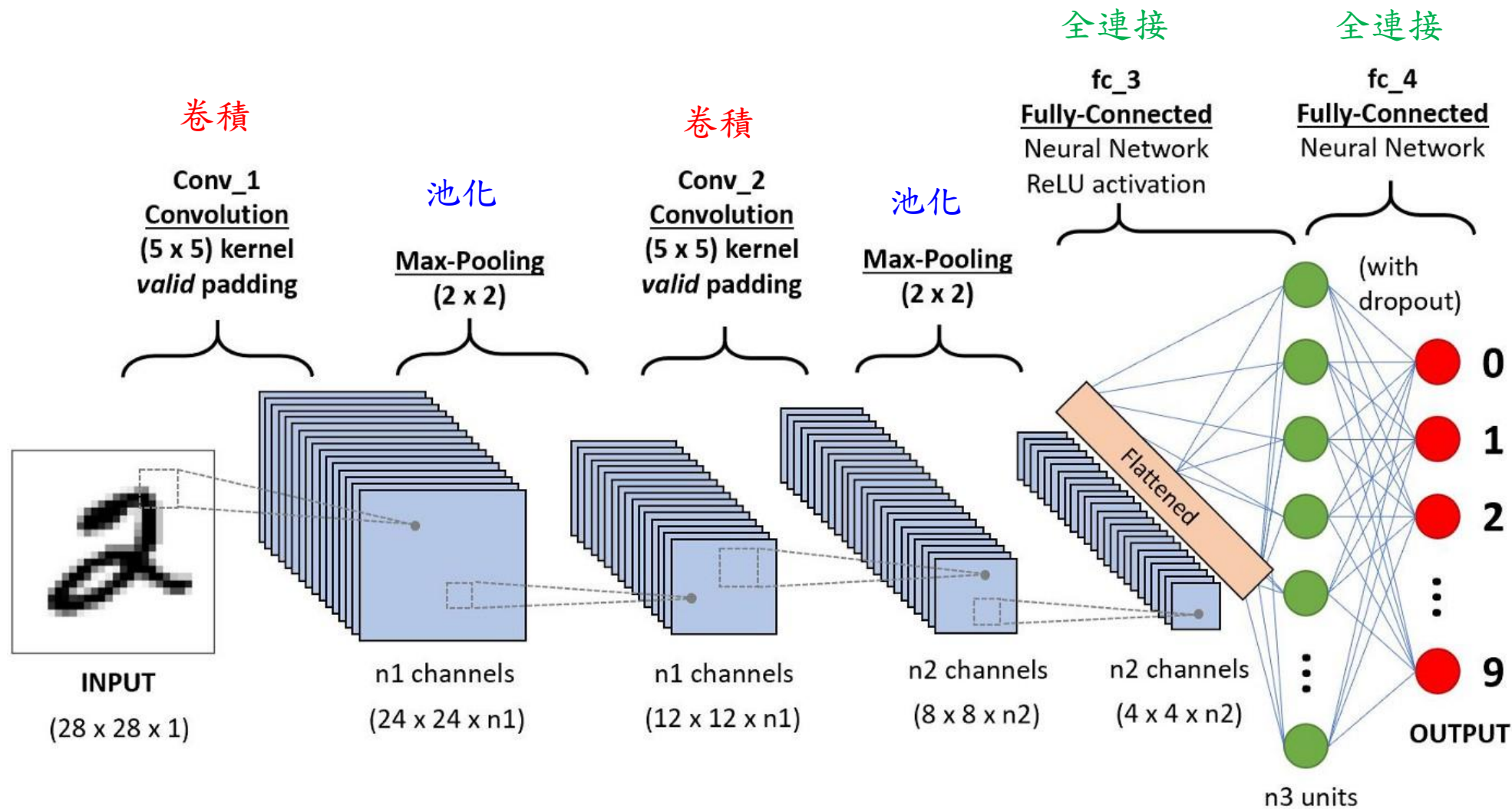
Pytorch 訓練網路

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6         data = data.to("cuda")
7         targets = targets.to("cuda")
8
9         # forward
10        scores = model(data)
11
12        loss = criterion(scores, targets) —————→ 算誤差
13
14        # backward
15        optimizer.zero_grad() —————→ 清空前一次累積的梯度
16        loss.backward() —————→ 根據loss算每個參數的梯度，並儲存
17
18        # gradient descent or adam step
19        optimizer.step() —————→ 做參數更新 SGD:  $w_1' = w_1 - \eta \nabla L(w_1)$ 
```


影像如何被計算

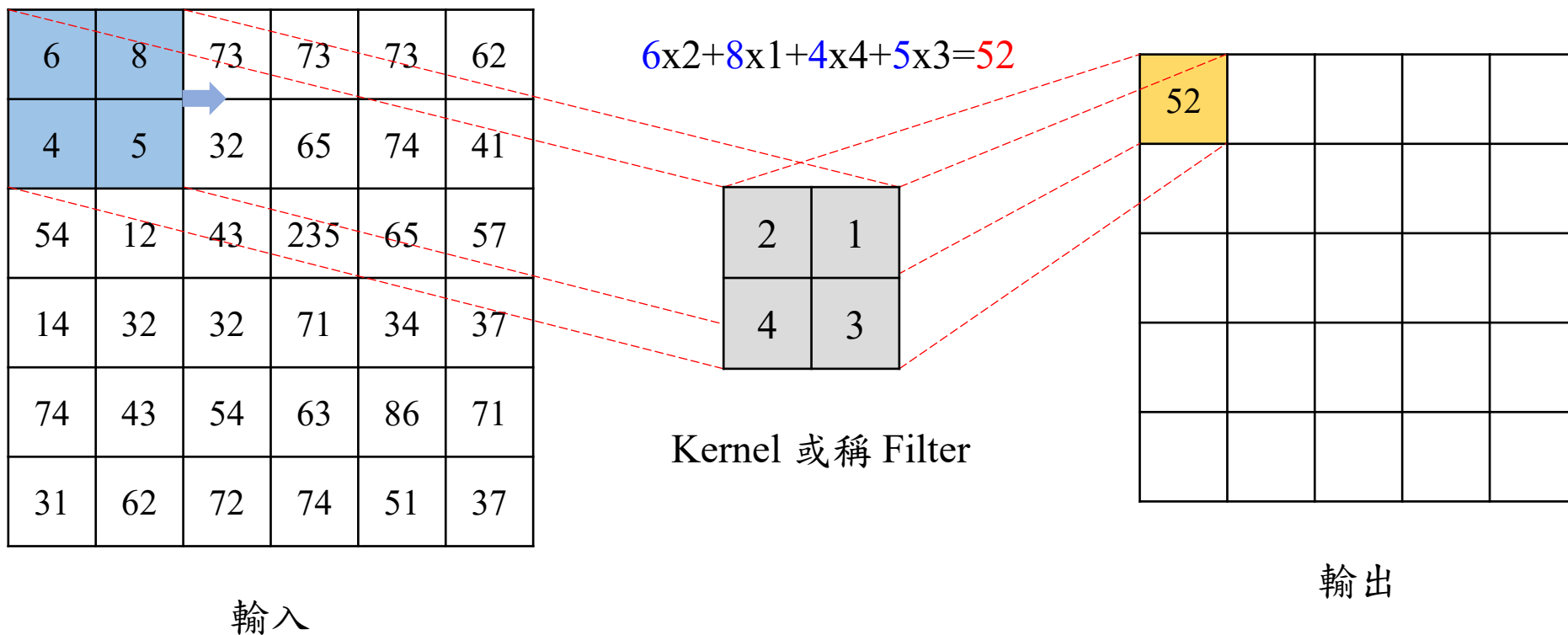


CNN簡介



卷積(Convolution)

透過kernel找尋特定特徵



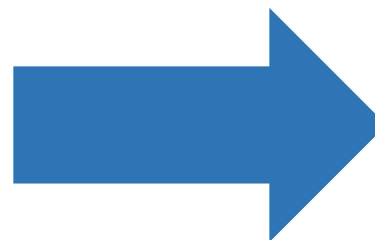
池化(Pooling)

取最大

231	32	65	74
123	43	235	65
32	32	71	34
43	54	63	86

池化前

挑選特徵最強的的地方，減少之後的運算量



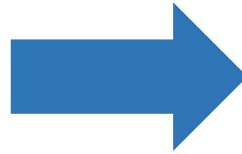
231	235
54	86

池化後

填充(Padding)

避免做卷積時步伐太大，直接撞牆，無法算到邊緣

231	32	65	74
123	43	235	65
32	32	71	34
43	54	63	86



0	0	0	0	0	0
0	231	32	65	74	0
0	123	43	235	65	0
0	32	32	71	34	0
0	43	54	63	86	0
0	0	0	0	0	0

Padding前

Padding後

CNN 權重更新

$$\begin{array}{ccc} \text{輸入} & \text{kernel} & \text{最終輸出} \\ \begin{bmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \\ x_{31} & x_{32} & x_{33} \end{bmatrix} \star \begin{bmatrix} w_{11} & w_{12} \\ w_{21} & w_{22} \end{bmatrix} = \begin{bmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{bmatrix} & \xrightarrow{\text{MaxPool}} & [z] \end{array}$$

每一個 y_{ij} 都是「對應的 2x2 區塊」跟 kernel 的內積：

$$\begin{aligned} y_{11} &= w_{11}x_{11} + w_{12}x_{12} + w_{21}x_{21} + w_{22}x_{22} \\ y_{12} &= w_{11}x_{12} + w_{12}x_{13} + w_{21}x_{22} + w_{22}x_{23} \\ y_{21} &= w_{11}x_{21} + w_{12}x_{22} + w_{21}x_{31} + w_{22}x_{32} \\ y_{22} &= w_{11}x_{22} + w_{12}x_{23} + w_{21}x_{32} + w_{22}x_{33} \end{aligned}$$

$$[z] = \text{MaxPool}(y_{11}, y_{12}, y_{21}, y_{22})$$

$$\text{MaxPool} \left(\begin{pmatrix} w_{11}x_{11} + w_{12}x_{12} + w_{21}x_{21} + w_{22}x_{22} \\ w_{11}x_{12} + w_{12}x_{13} + w_{21}x_{22} + w_{22}x_{23} \\ w_{11}x_{21} + w_{12}x_{22} + w_{21}x_{31} + w_{22}x_{32} \\ w_{11}x_{22} + w_{12}x_{23} + w_{21}x_{32} + w_{22}x_{33} \end{pmatrix} \right) = [z]$$

CNN w_{11} 權重更新

輸入

kernel

最終輸出

$$\begin{bmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \\ x_{31} & x_{32} & x_{33} \end{bmatrix} \star \begin{bmatrix} w_{11} & w_{12} \\ w_{21} & w_{22} \end{bmatrix} = \begin{bmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{bmatrix} \xrightarrow{MaxPool} z$$

輸入

初始化權重0.1

範例:

$$\begin{bmatrix} 3 & 1 & 3 \\ 1 & 1 & 3 \\ 3 & 3 & 3 \end{bmatrix} \star \begin{bmatrix} 0.1 & 0.1 \\ 0.1 & 0.1 \end{bmatrix} = \begin{bmatrix} 0.6 & 0.8 \\ 0.8 & 1.0 \end{bmatrix} \xrightarrow{MaxPool} 1 \quad \text{Target (t表示)} = 5$$

$$MaxPool \left(\begin{matrix} w_{11}x_{11} + w_{12}x_{12} + w_{21}x_{21} + w_{22}x_{22} \\ w_{11}x_{12} + w_{12}x_{13} + w_{21}x_{22} + w_{22}x_{23} \\ w_{11}x_{21} + w_{12}x_{22} + w_{21}x_{31} + w_{22}x_{32} \\ w_{11}x_{22} + w_{12}x_{23} + w_{21}x_{32} + w_{22}x_{33} \end{matrix} \right) = z$$

1. 算loss

$$\begin{array}{ccc} \text{輸入} & \text{kernel} & \text{最終輸出} \\ \begin{bmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \\ x_{31} & x_{32} & x_{33} \end{bmatrix} \star \begin{bmatrix} w_{11} & w_{12} \\ w_{21} & w_{22} \end{bmatrix} = \begin{bmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{bmatrix} & \xrightarrow{\text{MaxPool}} & z \end{array}$$

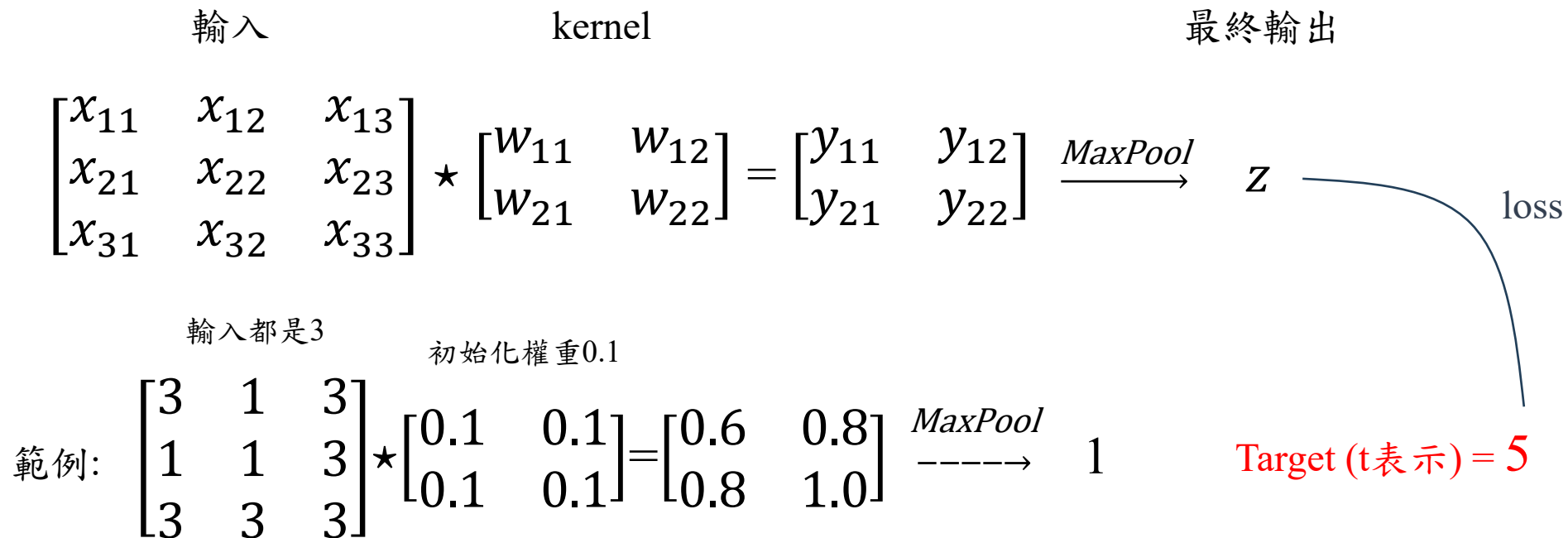
範例: $\begin{array}{ccc} \text{輸入都是3} & \text{初始化權重0.1} & \\ \begin{bmatrix} 3 & 1 & 3 \\ 1 & 1 & 3 \\ 3 & 3 & 3 \end{bmatrix} \star \begin{bmatrix} 0.1 & 0.1 \\ 0.1 & 0.1 \end{bmatrix} = \begin{bmatrix} 0.6 & 0.8 \\ 0.8 & 1.0 \end{bmatrix} & \xrightarrow{\text{MaxPool}} & 1 \end{array}$ Target (t表示) = 5

$$L = \frac{1}{2} (z - t)^2$$

$$\frac{\partial L}{\partial w_{11}} = \frac{\partial L}{\partial z} \cdot \frac{\partial z}{\partial w_{11}}$$

$$\frac{\partial L}{\partial z} = z - t$$

2. Loss 對 w_{11} 的微分



$$\frac{\partial L}{\partial w_{11}} = \frac{\partial L}{\partial z} \cdot \frac{\partial z}{\partial w_{11}}$$

$$\frac{\partial L}{\partial w_{11}} = \frac{\partial L}{\partial z} \left(\frac{\partial z}{\partial y_{11}} \frac{\partial y_{11}}{\partial w_{11}} + \frac{\partial z}{\partial y_{12}} \frac{\partial y_{12}}{\partial w_{11}} + \frac{\partial z}{\partial y_{21}} \frac{\partial y_{21}}{\partial w_{11}} + \frac{\partial z}{\partial y_{22}} \frac{\partial y_{22}}{\partial w_{11}} \right)$$

2. Maxpooling 倒傳遞 (關鍵)

$$\frac{\partial L}{\partial w_{11}} = \frac{\partial L}{\partial z} \left(\frac{\partial z}{\partial y_{11}} \frac{\partial y_{11}}{\partial w_{11}} + \frac{\partial z}{\partial y_{12}} \frac{\partial y_{12}}{\partial w_{11}} + \frac{\partial z}{\partial y_{21}} \frac{\partial y_{21}}{\partial w_{11}} + \frac{\partial z}{\partial y_{22}} \frac{\partial y_{22}}{\partial w_{11}} \right)$$

$$\begin{bmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{bmatrix} \xrightarrow{\text{MaxPool}} z$$

$$\frac{\partial L}{\partial w_{11}} = (z - t) \left(m_{11} \frac{\partial y_{11}}{\partial w_{11}} + m_{12} \frac{\partial y_{12}}{\partial w_{11}} + m_{21} \frac{\partial y_{21}}{\partial w_{11}} + m_{22} \frac{\partial y_{22}}{\partial w_{11}} \right)$$

這個 m 再向前傳遞時也會存起來

$$\begin{bmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{bmatrix} \odot \begin{bmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{bmatrix} = z$$

$$\begin{bmatrix} 0.6 & 0.8 \\ 0.8 & 1.0 \end{bmatrix} \odot \begin{bmatrix} 0 & 0.8 \\ 0 & 1 \end{bmatrix} = 1$$

$$m_{11}y_{11} + m_{12}y_{12} + m_{21}y_{21} + m_{22}y_{22} = z$$

3. y 對權重的微分

$$\frac{\partial L}{\partial w_{11}} = \frac{\partial L}{\partial z} \left(\frac{\partial z}{\partial y_{11}} \frac{\partial y_{11}}{\partial w_{11}} + \frac{\partial z}{\partial y_{12}} \frac{\partial y_{12}}{\partial w_{11}} + \frac{\partial z}{\partial y_{21}} \frac{\partial y_{21}}{\partial w_{11}} + \frac{\partial z}{\partial y_{22}} \frac{\partial y_{22}}{\partial w_{11}} \right)$$

$$\begin{aligned} y_{11} &= w_{11}x_{11} + w_{12}x_{12} + w_{21}x_{21} + w_{22}x_{22} \\ y_{12} &= w_{11}x_{12} + w_{12}x_{13} + w_{21}x_{22} + w_{22}x_{23} \\ y_{21} &= w_{11}x_{21} + w_{12}x_{22} + w_{21}x_{31} + w_{22}x_{32} \\ y_{22} &= w_{11}x_{22} + w_{12}x_{23} + w_{21}x_{32} + w_{22}x_{33} \end{aligned}$$

因此：

$$\frac{\partial y_{11}}{\partial w_{11}} = x_{11}, \frac{\partial y_{12}}{\partial w_{11}} = x_{12}, \frac{\partial y_{21}}{\partial w_{11}} = x_{21}, \frac{\partial y_{22}}{\partial w_{11}} = x_{22}$$

代回原式：

$$\frac{\partial L}{\partial w_{11}} = (z - t)(M_{11}x_{11} + M_{12}x_{12} + M_{21}x_{21} + M_{22}x_{22})$$

接著就可以 權重更新了

$$w_{new_11} = w_{11} - \eta * \frac{\partial L_{total}}{\partial w_{11}} \quad (\text{完})$$

參考資料

李弘毅-倒傳遞:

<https://www.youtube.com/watch?v=ibJpTrp5mcE>

印度小帥-倒傳遞

<https://www.youtube.com/watch?v=YO1OLxrMUOw&t=563s>

Andrej Karpathy-倒傳遞:

<https://www.youtube.com/watch?v=VMj-3S1tku0>

ChatGPT-倒傳遞:

有問必答 <https://chat.openai.com/chat>